BKI 114: Introduction Human-Computer Interaction 2006-07

Fitts' Law

task 2

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Introduction

This report is written for the subject Human Computer Interaction at the Radboud University Nijmegen in the Netherlands. Human Computer Interaction examines how a as good as possible interacton between human and computer can be achieved, in order to make the computer support the human in tasks as good and efficient as possible. The Human Computer Interaction subject has a very big scope and this paper will focus on a small part of the subject.

Often when humans interact with a computer, a keyboard and mouse are used to make the human tell it's wishes to a computer. This report will focus on a specific part of mouse tasks. When looking at a mouse pattern a user follows (different mouse pointer locations at different times and mouseclicks at different times) some regularity can be found. There seems to be a specific movement to a certain place on the screen followed by a click repetively, at least when drag and drop is being ignored.

Zooming in at this specific tasks, some properties become clear in every case which can thus be generalised. One travels from a source to target location. Both locations have a x and y value and with a little help from Pythagoras the shortest possible travelled distance can be calculated. Also when different time intervals the mouse pointer position is being stored, a very close approach of the actual travelled distance can be made. Another property is the time this task actually takes. Finally it's possible to look at the surface size of the target area. Often a user doesn't need to click on a specific pixel but somewhere in an area, a button by example.

Taking in consideration the above paragraph, 3 distinct variables can be defined, which in some way might be dependent from each other in besides them completely constant circumstances. These variables are the distance being travelled by mouse, the area of the target area and the time the specific task took.

This report will discuss current findings about the dependency of the variables and also introduce an experiment in which the current findings are being used and tested. Besides testing this, the report will also introduce something else variable and test wheter this can improve performance. When it can, it can have some positive impact in what way interfaces are being designed. Several hypotheses will make clear about what needs to be examined and what the vision about them is before the experiment. Finally the experiment's results will be derived and analyzed from the experiment. After this the hypotheses can be tested, and a conclusion about the succesfullness of the experiment can be drawn.

Fitts' law

In this chapter a discovery regarding pointing at something for humans (in this case using the mouse) will be discussed and Fitts' findings on it. First in order to provide some background information, Paul Fitts his background will be reviewed.

Paul Fitts (1912-1965) was a psychologist at Ohio State University. He developed a model of human movement. It turned out to be one of the best studied and most succesful mathematical models of humans motion. By focusing his attention on human factors during his time as Lieutenant Colonel in the US Air Force, Fitts became known as one of the pioneers in improving aviation safety. In 1965 he died unexpectedly at the age of 53. He received degrees in psychology. University of Tennessee (BS 1934) Brown University (MS 1936) University of Rochester (PhD 1938).

Let's go back to the theory. Wikipedia offers the following description of the model:

$$T = a + b\log_2\left(\frac{D}{W} + 1\right)$$

T is the average time taken to complete the movement. (Traditionally, researchers have used the symbol MT for this, to mean movement time.
a and b are empirical constants, and can be determined by fitting a straight line to measured data.
D is the distance from the starting point to the center of the target. (Traditionally, researchers have used the symbol A for this, to mean the amplitude of the movement.)
W is the width of the target measured along the axis of motion. W can also be thought of as the allowed error tolerance in the final position, since the final point of the motion must fall within ± W/2 of the target's centre.

Regarding the experiment some caution have to be made considering the following facts:

It applies only to movement in a single dimension and not to movement in two dimensions (though it is successfully extended to two dimensions in the Accot-Zhai steering law). In the later mentioned experiment this is being ignored and as one hypotheses will say, every dimension is equally hard.

It describes simple motor response of, say, the human hand, failing to account for software acceleration usually implemented for a mouse cursor. Unfortunately the software platform being used includes this acceleration and thus relatively better performance on tasks with a bigger distance might be expected.

It describes untrained movements, not movements that are executed after months or years of practice (though some argue that Fitts' law models behaviour that is so low level that extensive training doesn't make much difference). Every subject in the experiment has had extensive mouse training, but the behaviour can be considered very low level, which might account for justification to ignore this caution.

Considering the above information, it's possible to think of an experiment with some hypotheses that uses the application of Fitt's law. This experiment is being described in the next chapter.

The experiment

Reading the previous chapter of this report, it becomes challenging to set up an experiment in which Fitts' law is being applied or tested. Before even thinking of what to make or how to make something, it's important to make a list of hypotheses that are wished to be proven. These hypotheses are still very general, and after they've been listed a concrete implementation of how to prove them will follow. Six different hypotheses have been defined:

1) Vertical, diagonal and horizontal movement of the mouse are all equally hard
When a subject goes from object on the screen to another, they vary in x and y coordinate. Knowing the differences in coordinates, leads to an angle from 0 to 360 degrees in which the mouse pointer has to travel. In first case this angle is being ignored and being treaten equally to any other angle. Whenever results turn out to be very irregular, this phenomenon can be kept in mind. The reason to assume the angle being travelled not mattering is that every subject used an optical mouse. Unlike a mouse with 2 wheels and a ball, every direction of movement has the same resistance from the mouse. However, the angle of travel possibly mattering could also have cognitive reasons.
2) The time of tracing an object on the screen has the same relative difficulty index as having to click on an object on the screen
When every object that the subject has to click on on the screen appears randomly, some time needs to be taken by the subject in order to detect the next object. This time is not included in Fitts' law, but whenever this time is distributed in relatively the same way as the mouse task, it will the equation of Fitts' law, but only with higher constant values. Assuming the hypothese is true has the implication that one has to assume 2 other things, namely: whenever an object is twice as small, the difficulty of detecting it is twice as hard and whenever an object is twice as far away from the current mouse pointer, the difficulty of finding it is twice as hard. The first one seems to make sense, but the second one requires some more thinking. The second implication basically says that a user will have a local scope of viewing instead of seeing the whole picture.
3) Marking the center of an object different makes it easier for a subject to target it and thus will lower the difficulty index
Sometimes subjects have to click on objects that are relatively big and thus making the task easy. But a disadvantage of having to click big objects is that it's hard to focus on what's the middle of the object and possibly because of that hard to aim at the middle of the object. Why might aiming at the middle of the object offer any advantage? It'll give the subject a point to focus on which only should be approached. Especially when the object it's big it's hard to know where to aim, in order to have the highest likelyness to move into an object without even trying to exactly moving on to it.
4) Smaller objects cause more misclicks
While looking at different misclicks, the frequency they occur can be displayed in the size of the circle dimension. This produces a graph with a circle size on the x-axis and a relatively frequency in which a misclick occured. The above hypotheses says that this graph will show a bigger y value when the x value increases.
5) Misclicks are all missed in the same range, no matter the difficulty of the task
Occasionally, even though it's purposely not part of the initial task, a misclick is being made. One could wonder whether the distance of the misclick depends on the size of the circle or the distance being travelled or both combined. This leads to a graph in which the difficulty is plotted against the distance of the misclicks. According to the hypotheses it should lead to a constant distance of misclick, no matter what the difficulty of the task may be.

After all these hypotheses have been defined, an experiment can be made up, in which they can be tested in an emperical way, as shown in next chapter.

The task

In order to test the previously listed 5 hypotheses a generic task needs to be defined that can test them all in an accurate and reliable way. Before the task was being thought of, some parameters were already known. The task can be applied on 10 people which either have 3-5 mins of time to involve in the experiment. Also it was known that the task had to be performed on a computer, preferably using the mouse or another pointing device (which can be derived from the definition of Fitts' law).

Considering everything mentioned so far, the task turned out this way: Every subject has to perform the same task, so the experiment is within subjects. Every subject takes place in front of a computer screen and only has the mouse to perform actions. After starting up the software program for the task, an empty canvas appears. When the subject clicks on this canvas, the actual task starts. A circle appears on which the subject has to click. After the subject does that, another one appears. This goes on for the time the experiment lasts. Every circles has some parameters that are generated randomly within some limits. Those parameters are:

- Location: a x and a y coordinate that is somewhere on the (viewable) part of the canvas. This is where the circle is drawn and thus visible to the subject.
- Size: the radius of the circle in pixels. This is the size of the circle the user sees.
- Incircle: a boolean. Whenever this boolean is true, another circle with half the radius of the initial one is drawn on top of original circle.

Besides the parameters of the different circles that are stored, some other information is being gathered during the experiment. The information is being written to a logfile at every mouseclick:

- Location: The location of the mouse pointer, when the mouse button was being clicked.
- Timestamp: The actual time in miliseconds since the start of the software application
- ClickOnCircle: A boolean telling whether the mouse click was on the circle or not
- Distance: The distance between the current and previous circle

Inspecting the log file leads to all the information needed in order to test the hypotheses.

To get slightly more information about the task, appendix A with the log book and appendix B with the screenshots are worth giving a look.

Analyzing the data

While examining the log file, some graphs have been produced, shown in appendix C. These graphs make a better visual representation for humans above have just numbers to look at. All the gathered information has been used to test the different hypotheses, which is done in this chapter.

The first hypotheses has been defined because Fitts' law only supports 1 dimensional movements, while the task performed by the subject involved two dimensions. In the calculation of the difficulty index the distance between the two circles has been used and thus the 2 dimensions have been converted to one using pythagoras. A way to test wheiter this is a legitimate thing to do, is by plotting a graph with a performance (difficulty/time) index on one axis and movement direction (x movement-y movement) on the other. Of course this graph should only be made per subject, in order to have a constant performance potential. In appendix C this graph is plotted named 'performance from vertical to horizontal'. As can be derived from the plot, there is no corellation between the both axis and thus can one assume the hypotheses has been proven to be true.

Hypotheses number 2, saying that the time of tracing an object on the screen has the same relative difficulty index as having to click on an object on the screen, can be checked by seeing if Fitts' law can be applied to this experiment. When this is not (clearly) the case, it'll mean that the time to trace an object has it's own difficulty index, which depends on different variables or perhaps the same variables but different importance or relation. Looking at the 10 user graphs in appendix C containing the results of the different subject, makes clear that for every subject on average the time gets bigger whenever a task becomes more difficult, but looking at the dots only, it becomes clear that there must be some external factors, that affected the results of the experiment, since all the dots aren't in a straight line. These external factors could be user dependant such as boredom, distraction, annoyment, exhaustion, but considering the time of the experiment and also keeping in mind that every subject wants to score well in an experiment, it should be possible to ignore these factors. Especially when a bigger group of subjects is being used. After considering the previously mentioned and assuming Fitts' law is right, the only factor that could've caused the results not to be as expected, is the cognition time of tracing an object and making up a task how to be able to click it and thus for estimating the time to trace an object, Fitts' law is not applicable and this hypotheses is false.

Hypotheses number 3, stating that marking the center of an object with a color improves performance is something that can be tested by looking at the ten results graphs of the different subjects named 'Results subject X' in appendix C of this document. The red dots represent the clicks without the extra circle in the middle and the cyan dots with. Besides the dots corelation lines have been drawn, representing the correlation of the dots. Looking at different subjects a slightly different correlation can been seen, but not significant. Also noticable is that for different subjects, a different method has better scores and thus it's tempting to say that the differences are caused by either external factors (concentration, boredom etc.) or by the way the difficulty of the task is being defined (equal weights for distance and size). This leads to the conclusion that no remarkable difference can be indicated.

Hypotheses number 4, saying that smaller objects cause more mistakes can be derived from the data, by taking the average and median value of the circle size and compare these to an expected average if there wouldn't be correlation between them. The results, in which all subjects are included, are as following:

Average radius of circle presented to subject: 35
Average radius of misclicked circles: 22,87
Median value of misclicked circles: 17

These differences seem relatively big and considering the number of misclicks that have been included in the measuring it's safe to say that relatively more mistakes are made on smaller objects.

Finally hypotheses number 5, suggesting that the distance of a misclick doesn't depend on the difficulty of its task. A graph assisting in this has been calculated and included in Appendix C of this paper named 'Distance of misclicks in certain difficulty'. Something remarkable is noticable in this graph while looking at the regression line. It seems that the harder a task is, the more accurate the user is leading to relatively smaller distances of mistakes. Although the data doesn't show the reason of this result, one can imagine that overestimating might be a cause. When users are challenged with a task that seems easy, they might overestimated their performance and as a result actually make big mistakes. It's important to keep in mind thought, that the amount of data to investigate this is relatively small, and that this hypotheses needs more testing in order to deliver accurate results.

Conclusion

An experiment has been made in order to test 5 hypotheses strongly related to Fitt's law. This experiment has been done by different subjects, and the results were used in order to draw a conclusion. The results of the hypotheses can be structurally listed in the following way:

#	hypotheses	true	remark
1	Vertical and horizontal movement is equal	Yes	No performance difference can be seen in different tasks with different moving directions
2	Detecting an object is equally hard to performing the task	No	Fitts' law didn't turn out to be true for the task
3	Marking a target of an object improves performance	No	Log book data didn't show a remarkable difference
4	Smaller objects cause more misclicks	Yes	The average and median size of the objects that were misclicked was smaller then average
5	Misclicks are in the same range, no matter the difficulty of the task	Yes	There wasn't a big difference, although harder tasks seem to have smaller misclick distance

Considering the above results some improvements could have been made to make the experiment more successful:

- Static direction of movement, in order to reduce or completely minimize the cognition time of the task
- Several the same tasks following each other up, in order to have a stable average on the performance of a specific task

Although the whole experiment hasn't turned out to be completely useful, some valuable information can be derived from it. 3 hypotheses have proven to be true and with 2 other hypotheses it's clear that the countrary of what was stated turned out to be true.

Appendix A: Log book sample

A sample of the logbook as generator by the program:

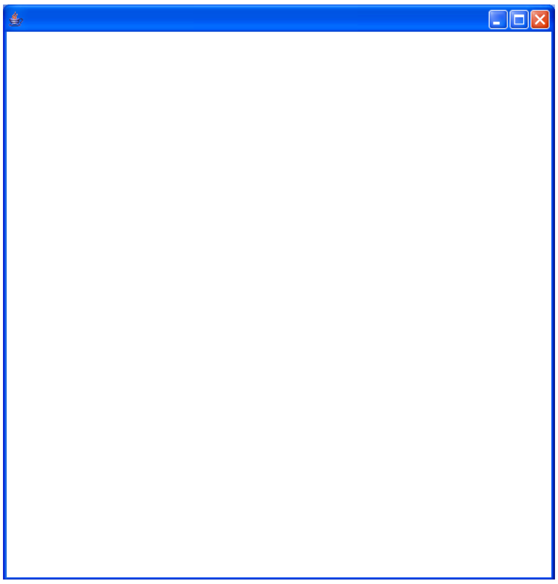
1176292670109	10609	455	434	488	444	376	51	10	1
1176292670109	11187	351	447	345	428	143	50	10	1
1176292670109	11750	76	256	80	258	314	15	10	1
1176292670109	12422	374	246	402	266	322	52	10	0
1176292670109	13094	390	347	386	349	84	23	10	0
1176292670109	13656	64	445	82	432	315	47	10	1
1176292670109	14328	154	145	143	144	294	36	10	1
1176292670109	14828	234	134	238	141	95	43	10	0
1176292670109	15437	340	147	341	154	103	24	10	0
1176292670109	16406	261	458	268	460	314	8	10	0
1176292670109	17094	136	191	140	201	288	20	10	0
1176292670109	17594	333	536	314	482	330	31	9	0
1176292670109	18000	283	470	266	467	50	57	10	0
1176292670109	18609	134	380	128	373	166	62	10	1
1176292670109	19094	338	299	313	295	200	55	10	1
1176292670109	19516	123	298	130	308	183	54	10	1
1176292670109	20047	156	330	98	322	34	64	10	1
1176292670109	20594	357	332	440	336	342	60	9	0
1176292670109	21125	284	55	279	80	302	54	10	0
1176292670109	21766	400	228	404	236	199	27	10	1
1176292670109	22312	457	223	443	230	39	47	10	1
1176292670109	23031	230	113	168	125	294	24	9	0
1176292670109	23578	233	391	256	468	354	55	9	0
1176292670109	24625	446	312	428	309	234	32	10	0
1176292670109	25156	316	509	320	508	226	33	10	0
1176292670109	25781	411	374	512	366	238	31	9	0
1176292670109	26437	68	127	143	170	417	56	9	0
1176292670109	27453	96	328	120	329	160	21	9	0
1176292670109	28453	530	537	502	498	417	53	10	0
1176292670109	29391	205	483	183	491	319	15	9	0
1176292670109	29891	140	380	141	400	100	39	10	0
1176292670109	30719	171	185	169	179	222	9	10	0
1176292670109	31719	445	277	470	281	317	31	10	0
1176292670109	32328	165	365	162	361	318	50	10	0
1176292670109	33187	285	281	296	283	155	9	9	1
1176292670109	33844	338	346	341	353	83	26	10	1
1176292670109	34594	112	233	114	231	257	11	10	1
1176292670109	35156	243	89	248	91	193	59	10	0
1176292670109	36312	284	279	326	273	198	20	9	0
1176292670109	37031	504	468	516	468	272	29	10	1
1176292670109	37937	322	314	318	302	258	36	10	1
1176292670109	38969	513	436	507	435	231	8	10	0
1176292670109	39922	114	340	112	346	404	27	10	1
1176292670109	40875	466	219	473	221	382	7	9	1
1176292670109	41516	130	439	115	454	427	62	10	1

Every line in the logbook has been generated after a mouse click. The meaning of the different column from left to right are:

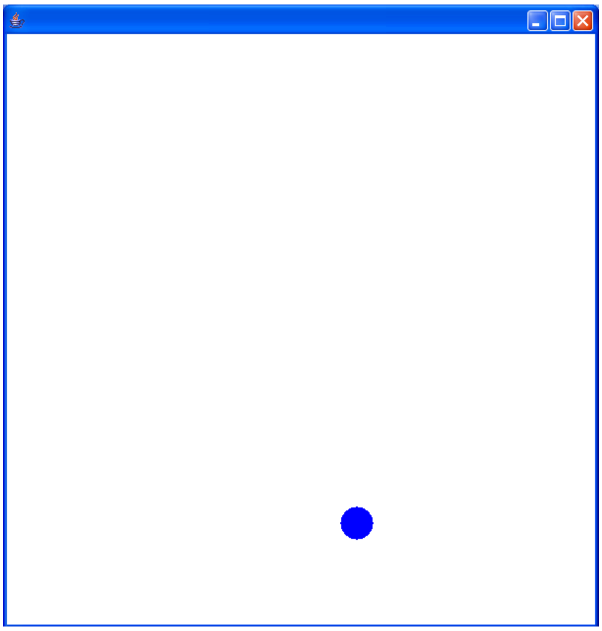
- unique user id (the timestamp of when the program started)
- session time stamp (the elapsed time in the program session)
- mouseX, mouseY (the position of the mouse when the subject clicked)
- circleX, circleY (the position of the middle of the target circle)
- distance (the distance of the target circle compared to the previous target circle)
- radius (the size of the target circle)
- miss? (10 when clicked on the target circle, 9 when missed)
- Incircle (1 when the circle included a target circle, 0 otherwise)

Appendix B: Screenshots

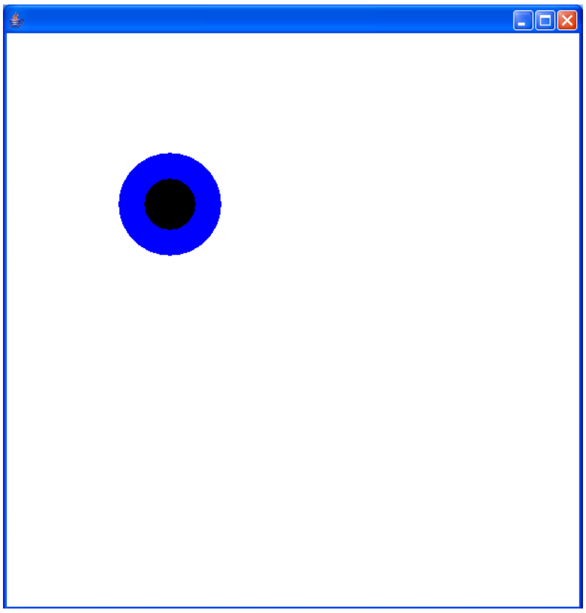
In order to give an impression of the task the subjects performaned, some screen shots have been included.



This is the first the subject sees. After making a first (random) mouse click the task starts.



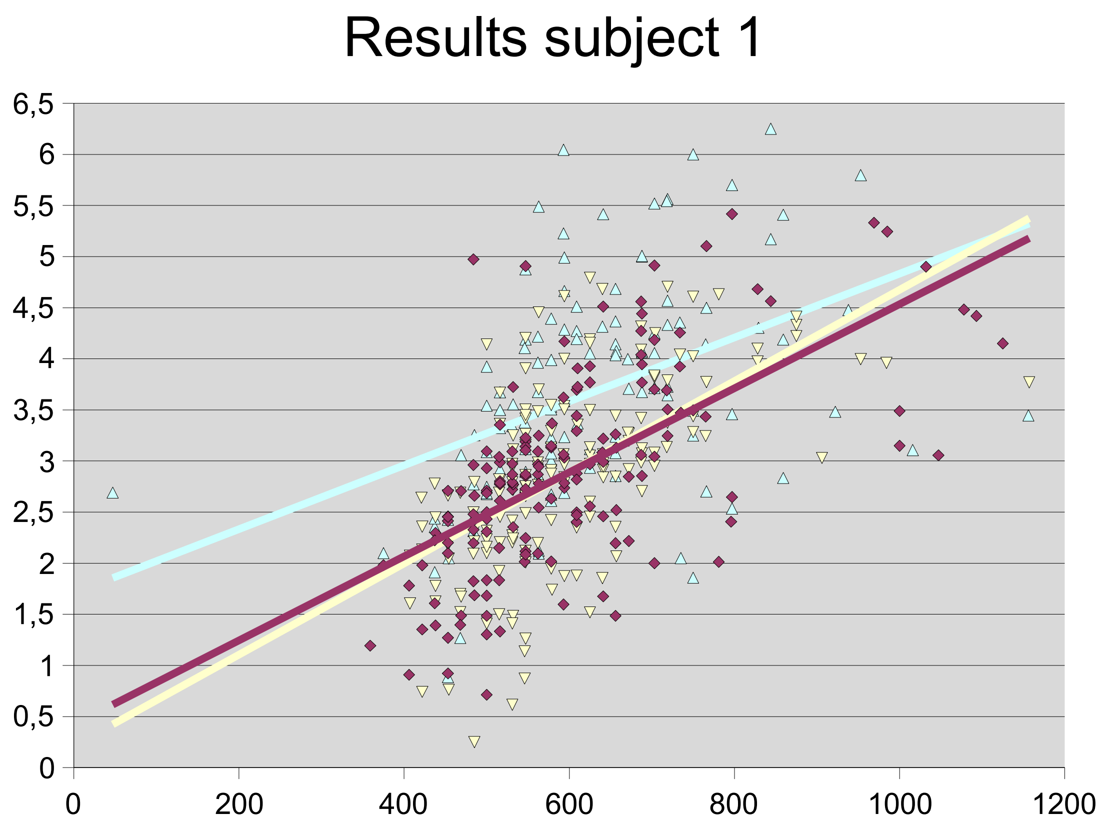
A circle with random size and location on the canvas. The user has to click on it as quickly as possible.

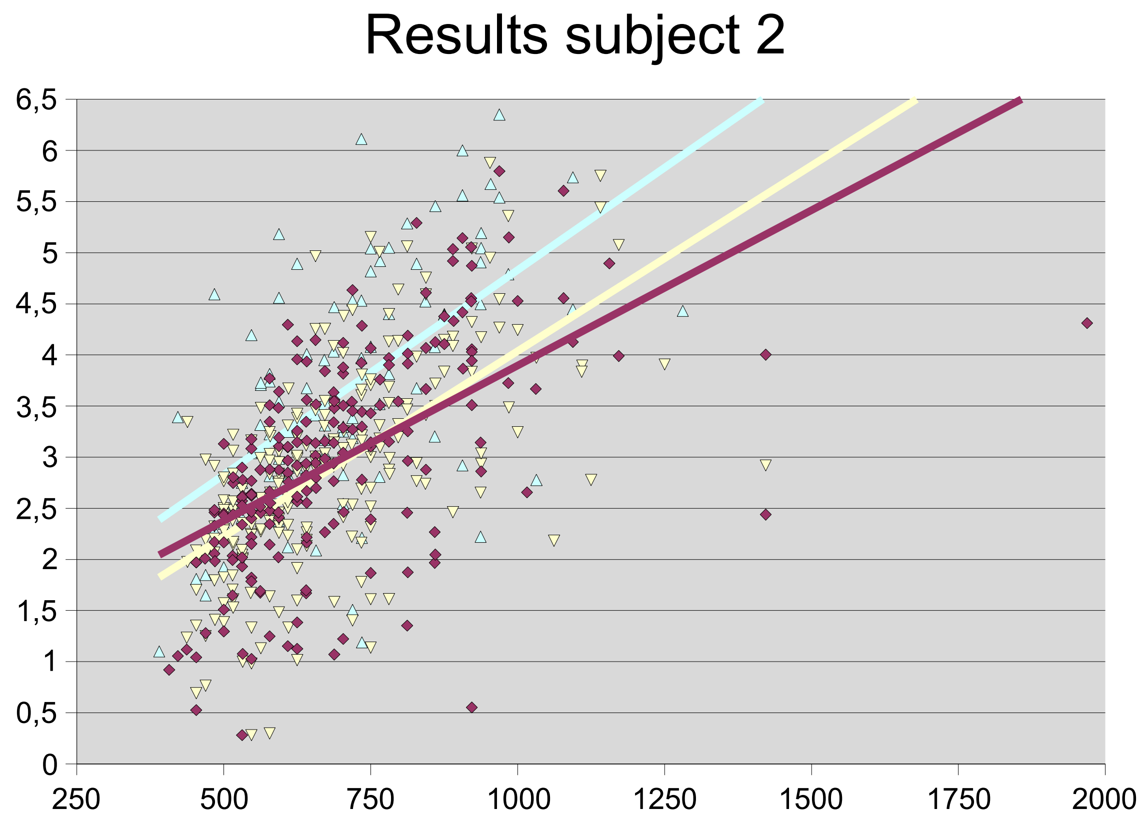


A circle with an extra target circle (50% radius of the bigger circle) with random size and location on the canvas. The user has to click on it as quickly as possible.

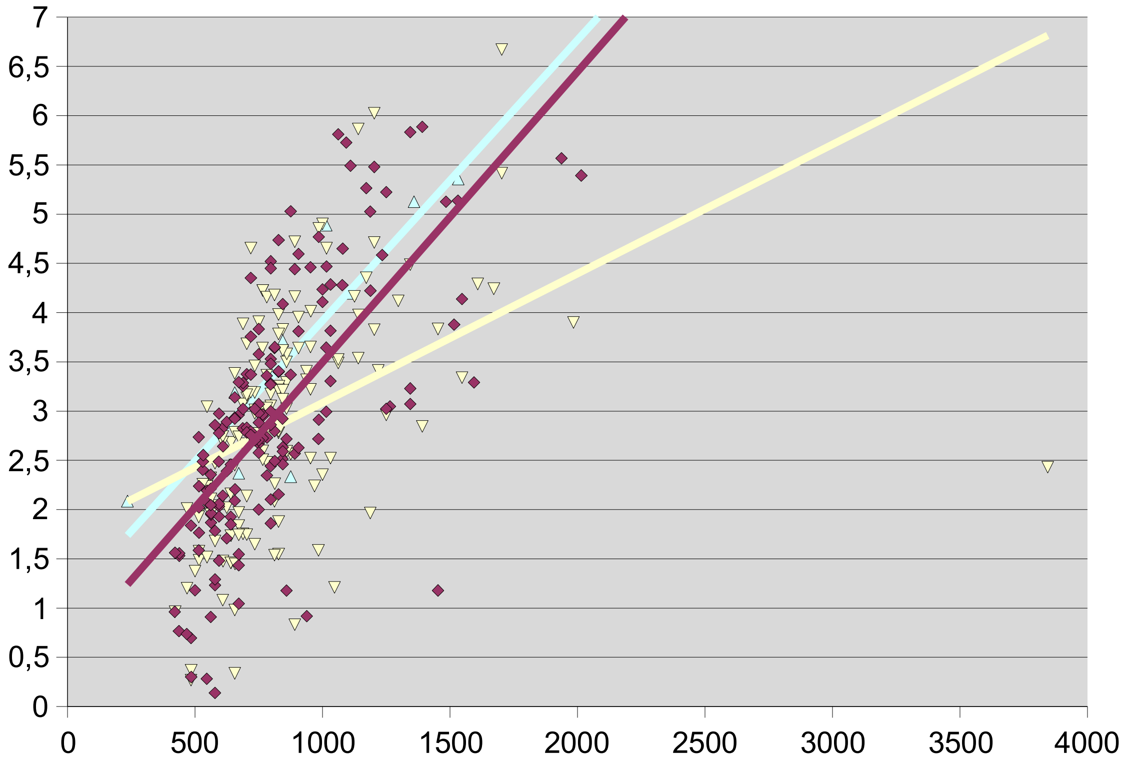
Appendix C: Several Graphs

10 graphs of different subjects. On the vertical axis a difficulty index has been shown (as stated by Fitts) and on the horizontal axis the time in miliseconds.

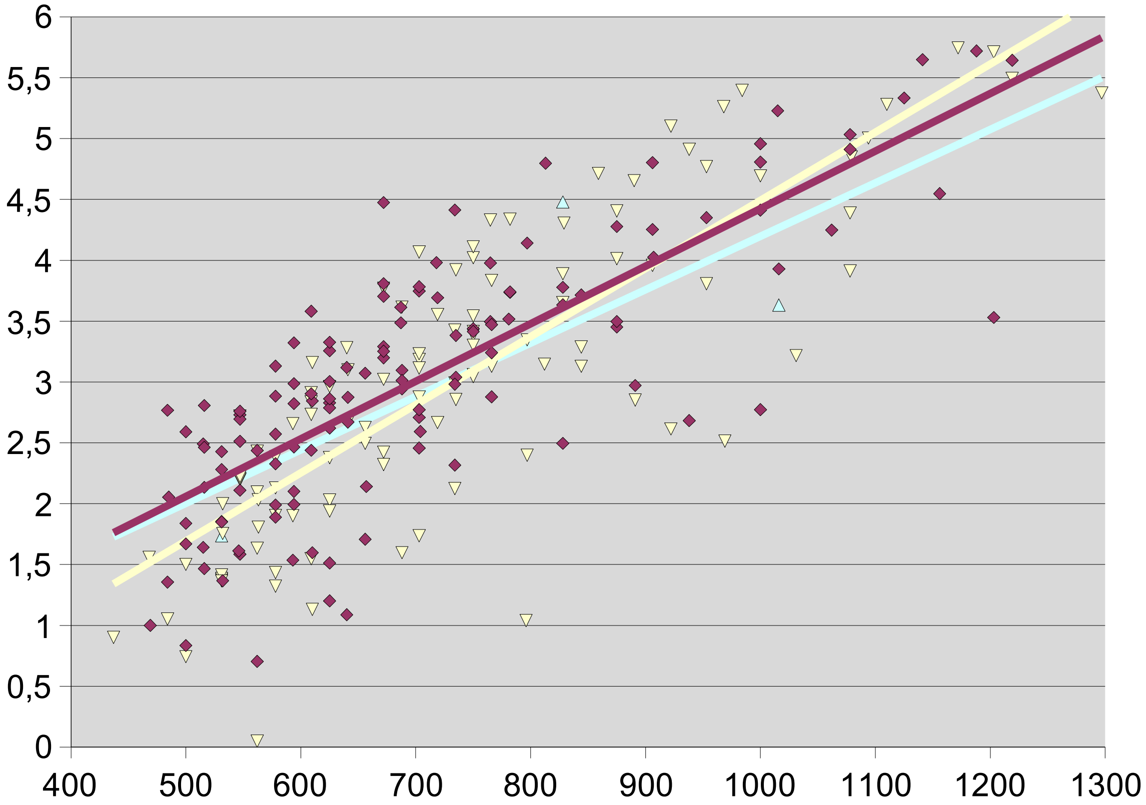


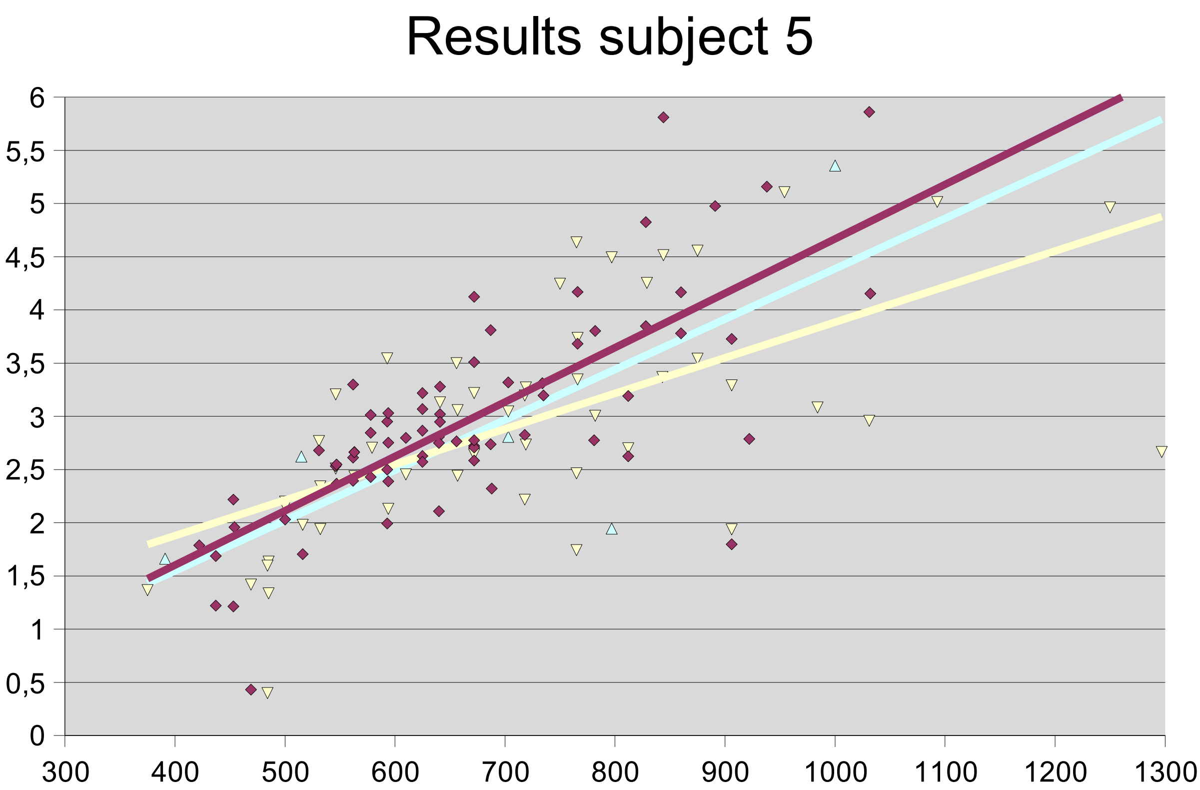


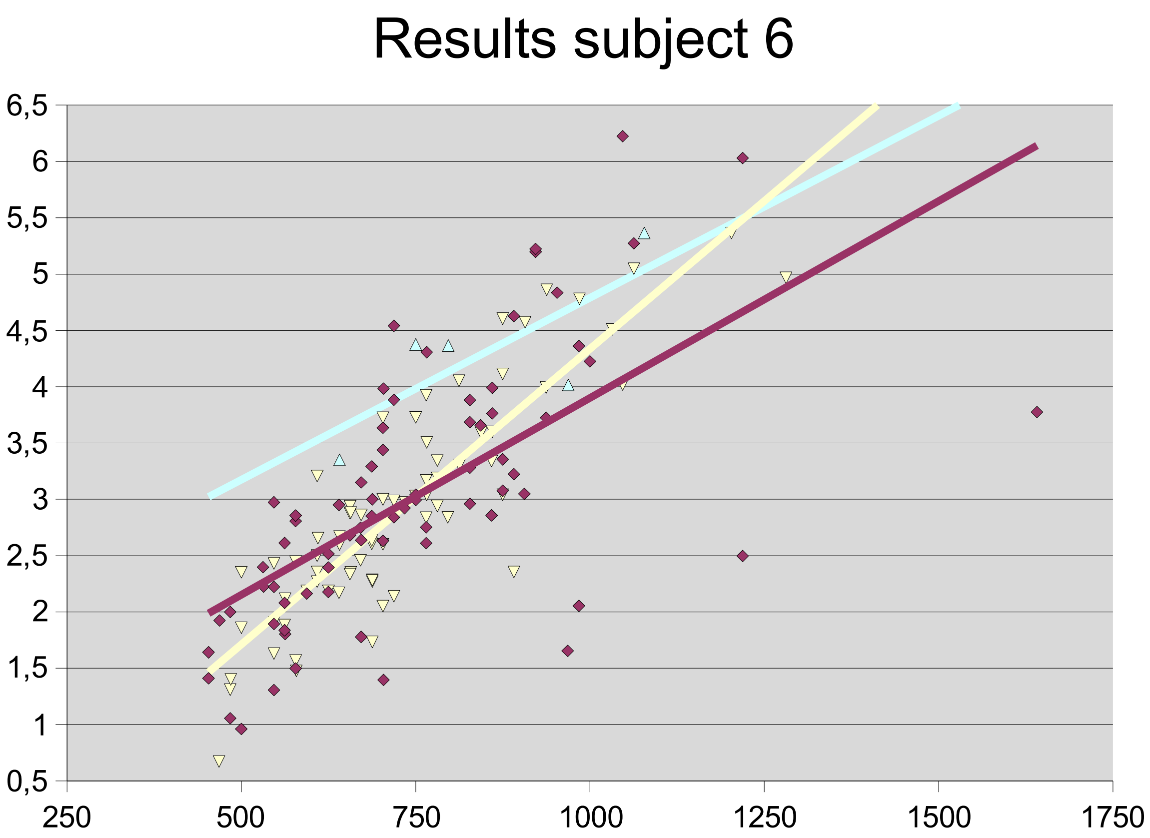
Results subject 3



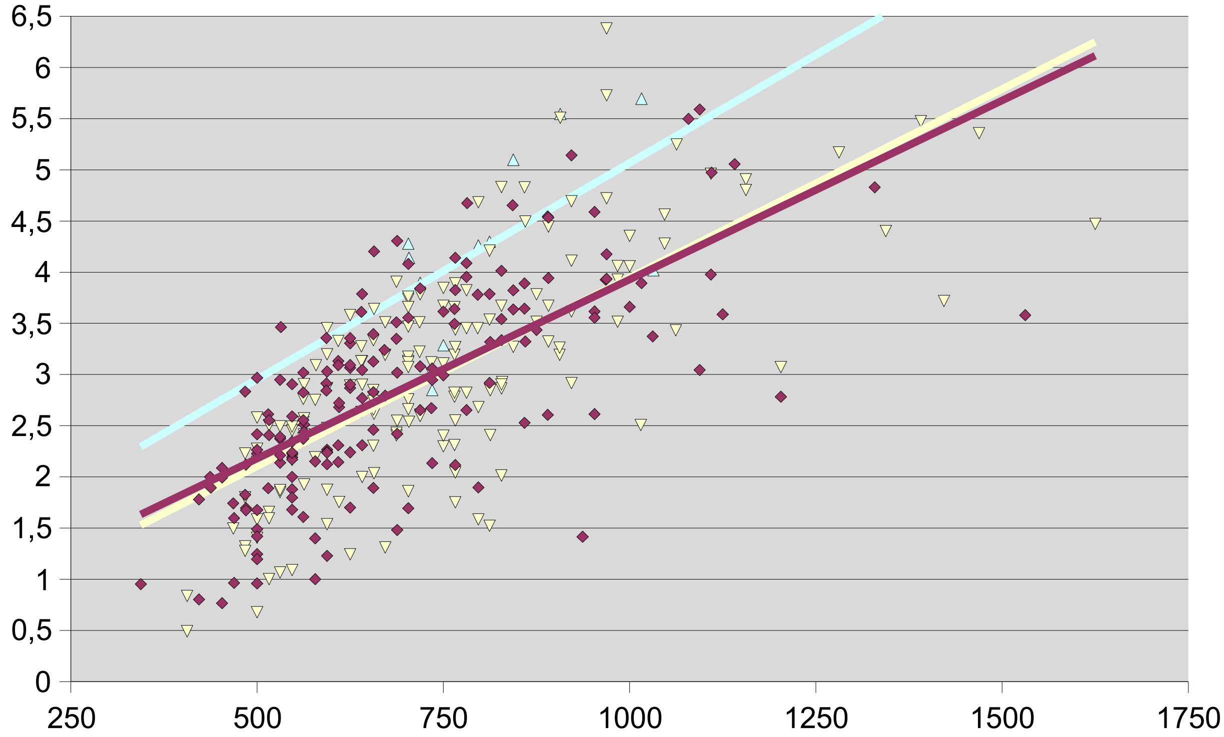
Results subject 4



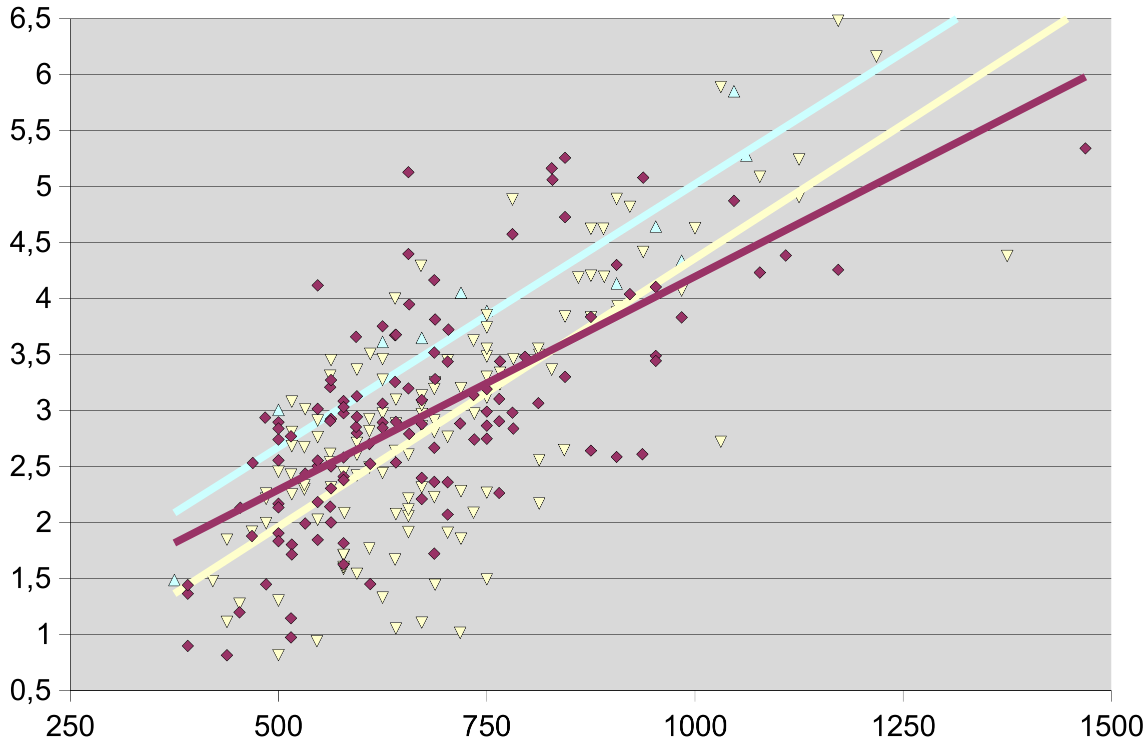

Results subject 5
6
5,5
5
4,5
4
3,5
3
2,5
2
1,5
1
0,5
0
300
400
500
600
700
800
900
1000
1100
1200
1300

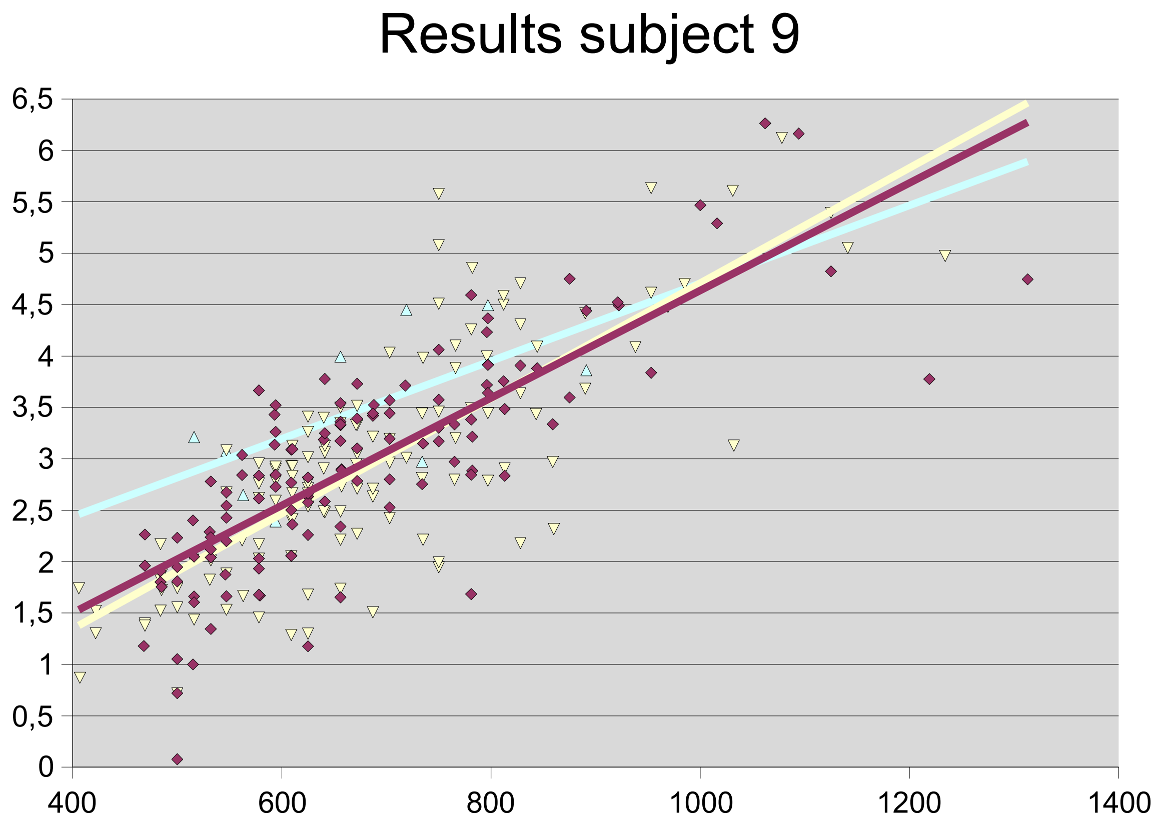

Results subject 6
6,5
6
5,5
5
4,5
4
3,5
3
2,5
2
1,5
1
0,5
250
500
750
1000
1250
1500
1750

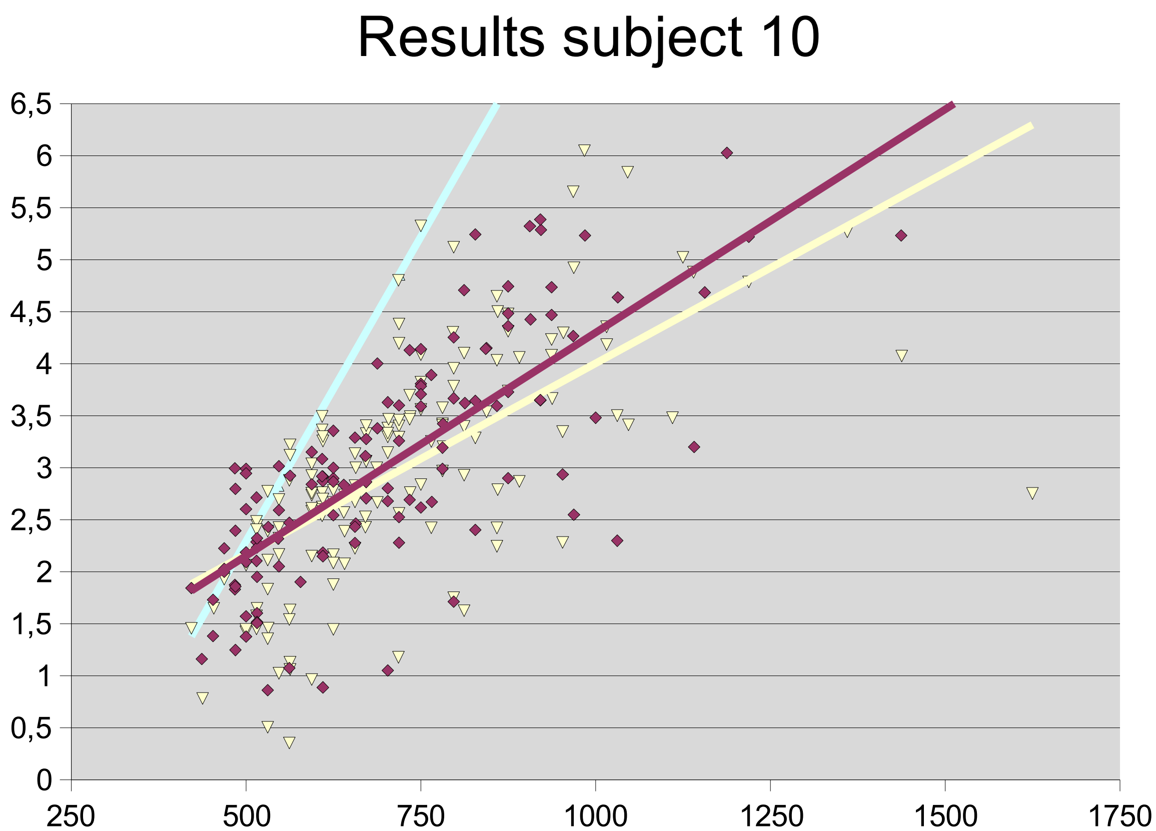
Results subject 7



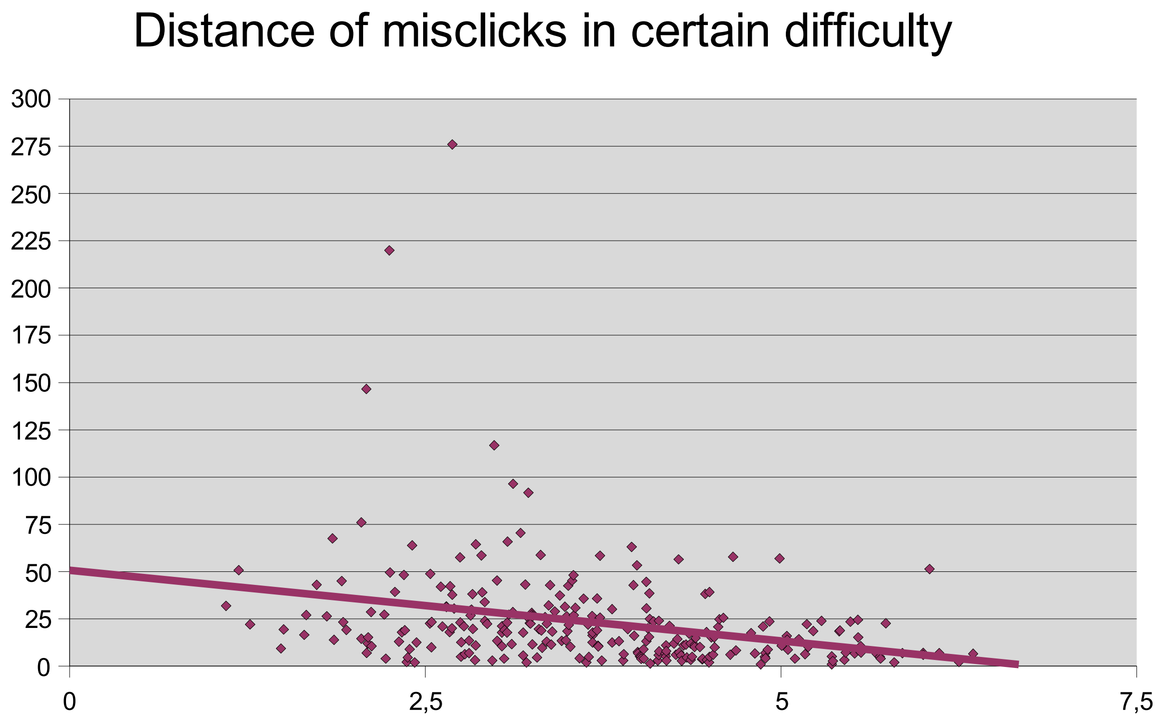
Results subject 8



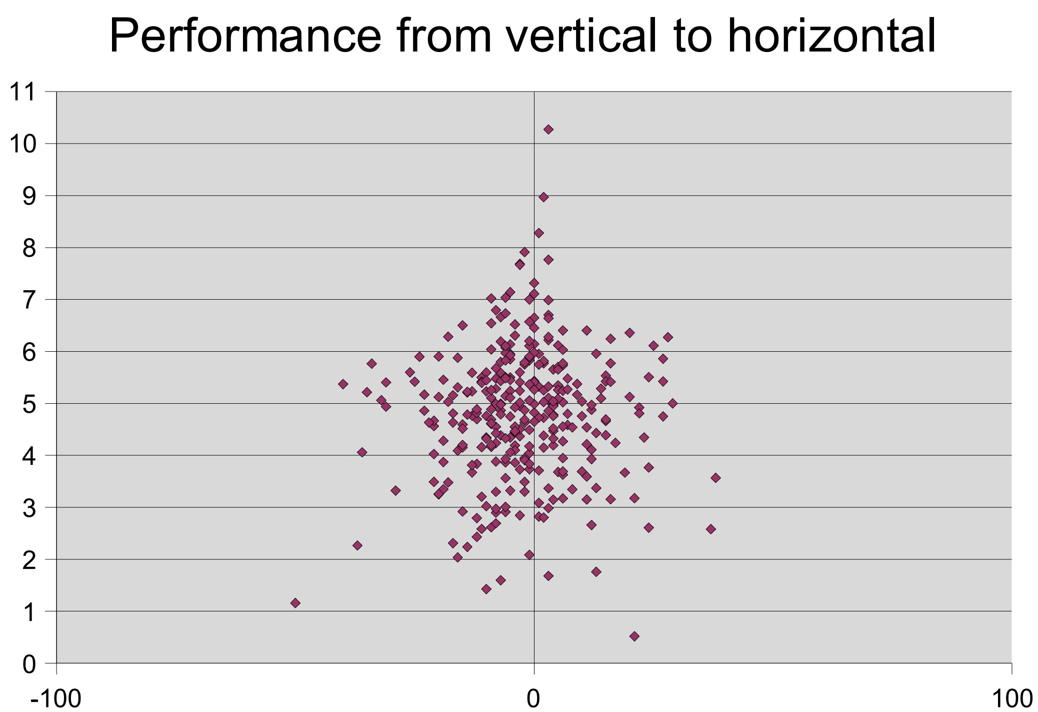

Results subject 9
6,5
6
5,5
5
4,5
4
3,5
3
2,5
2
1,5
1
0,5
0
400
600
800
1000
1200
1400


Results subject 10
6,5
6
5,5
5
4,5
4
3,5
3
2,5
2
1,5
1
0,5
0
250
500
750
1000
1250
1500
1750

The next graph shows the distance of a mistake (actually distance between mouse pointer and the very edge of the target circle on the vertical axis and the difficulty of the current task in which the misclick was made on the horizontal axis. The line shows the regression of the dots.



The following graph shows the difference in performance between horizontal and vertical movement. The x-axis contains the x movement compared to the y movement (deltax-deltay) and the y-axis shows the performance in that context (=difficulty/time). More left in the graph the more vertical movements (x<0) are shown and more right the more horizontal.



Appendix D: JAVA programming code

This appendix includes the code that has been used to test the different subjects. The code produces a log book in CSV format, which has been investigated using Open Office Spreadsheet.
In order to reduce the size of the code as much as possible, only one class has been used for the whole program, which is listed on this and the next page:

```
import java.awt.*;
import java.awt.event.*;
import java.util.*;
import javax.swing.*;
import java.io.*;
import java.lang.Math;

public class gui extends Canvas implements MouseListener {

        public static final int WIDTH = 600;
        public static final int HEIGHT = 600;
        public static final Dimension PREFERRED_SIZE = new Dimension(WIDTH,HEIGHT);
        public static int beginX, beginY, eindX, eindY, straal;
        public static boolean incircle;
        public static long begintijd;
        public static Canvas canvas;

        public void paint(Graphics g) {
                g.setColor(Color.white);
                g.fillRect(0,0,WIDTH,HEIGHT);
                g.setColor(Color.blue);
                g.fillOval(eindX-straal, eindY-straal, 2*straal, 2*straal);
                if (incircle)
                {
                  g.setColor(Color.black);
                  g.fillOval(eindX-(straal/2), eindY-(straal/2), straal, straal);
                }
        }

        public Dimension getPreferredSize() {
                return PREFERRED_SIZE;
        }

        private void veranderFiguur()
        {
                beginX=eindX;
                beginY=eindY;
                Random generator = new Random();
                incircle=generator.nextBoolean();
                straal = 5+generator.nextInt( 60 );
                eindX = 80+generator.nextInt( HEIGHT-160 );
                eindY = 80+generator.nextInt( WIDTH-160 );
                canvas.repaint();
        }

        public static void main(String[] args) {
               incircle=false;
               begintijd=System.currentTimeMillis();
               JFrame frame = new JFrame();
               frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
               Container contentPane = frame.getContentPane();
               canvas=new gui();
               MouseListener muisluisteraar= new gui();
               contentPane.add(canvas,BorderLayout.NORTH);
               frame.pack();
               frame.setVisible(true);
               canvas.addMouseListener(muisluisteraar);
        }
```

```
public void mousePressed(MouseEvent e) {
        try {
                FileOutputStream file = new FileOutputStream("log.txt", true);
                DataOutputStream out = new DataOutputStream(file);

                int goedeclick;
                int afstand=(int)Math.sqrt(Math.pow(beginX-eindX,2)+Math.pow(beginY-eindY,2));
                  int verschilX=e.getX()-eindX;
                  int verschilY=e.getY()-eindY;
                if ((verschilX*verschilX)+(verschilY*verschilY)<(straal*straal))
                   goedeclick=10;
                else
                   goedeclick=9;


                out.writeBytes(String.valueOf(begintijd));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(System.currentTimeMillis()-begintijd));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(e.getX()));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(e.getY()));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(eindX));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(eindY));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(afstand));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(straal));
                out.writeBytes(",");
                out.writeBytes(String.valueOf(goedeclick));
                out.writeBytes(",");
                if (incircle)
                        out.writeBytes("1");
                else
                        out.writeBytes("0");
                out.writeBytes("\n");
        }
        catch (IOException q) {
                  q.printStackTrace();
        }
        veranderFiguur();
       }

      public void mouseReleased(MouseEvent e) {}
      public void mouseEntered(MouseEvent e) {}
      public void mouseExited(MouseEvent e) {}
      public void mouseClicked(MouseEvent e) {}

}
```

Sources

http://www.ai.ru.nl/aicourses/bki114
BKI 114: Introduction Human-Computer Interaction 2006-07 Radboud University Nijmegen website

http://en.wikipedia.org/wiki/Fitts'_law
Fitts' law from Wikipedia, the free encyclopedia

http://en.wikipedia.org/wiki/Paul_Fitts
Paul Fitts from Wikipedia, the free encyclopedia